
PI Research Fellow Proposals

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1 Summary

Below are a series of short proposals and descriptions for experiments and research products that could be run through the PI Fellows program. Ideal targeted start date would be towards the beginning of May after the NeurIPS 2026 deadline, and happy to discuss or modify projects further to more closely align with Prime Intellect's incentives.

2 Relevance and impact statement

My biggest research interests focus on the idea of developing a more dynamically robust model architecture that's able to understand two key principles:

- (i) The idea that an action produces a consequence and, given a general objective function for the model, these action-consequence tuples carry useful information that can inform current and further learning; for more in-depth writing and some general formalisations, I've written blogposts already (linked here for action-consequence and here for associative learning).
- (ii) The idea that models should understand that they exist in the future past the current timestep that they are being evaluated on; specifically, if models don't understand that they exist on a larger time-horizon than the intermediate reward timeline, they have no ability to plan that far (this is also a concept I've written about here, inspired by Zafar and the Social Physics Lab).

Exploration in both of these ideas is admittedly more foundational and not directly related to the democratisation of LLM-based training procedures and architectures that Prime Intellect's agentic philosophy, PI Lab, and the PI protocol are primarily focussed on. However, I believe that these hypotheses are truly worth exploring and that, if done right, steps taken in these areas will inform further development of models that will either directly bolster or even potentially augment/outstrip agentic workflows today. Furthermore, the immediate next actions and explorations can also be done on LLM-based architectures and thus directly on Lab or the PI Training Console. More in-depth proposals are below:

3 Proposal #1: Direct development of an action-consequence learning algorithm or model through game-based systems

Objective: develop a model that's able to accurately create latent understandings or representations of the actions and consequences within a gamelike environment and use those to elicit strong performance on games when compared with traditional SOTA approaches like AlphaZero.

Progression:

- (i) Baseline post-training runs on standard open-source LLMs to determine whether action-consequence understanding in reasoning traces improves performance
- (ii) Explore RNN/SSM/dynamical systems-based models to effectively be able to predict actions and consequences
- (iii) Expand a "planning cortex" that's able to take in a representation of a given state within a game (think CNNs or encoded representations for Chess/Go, etc.) and then predict and plan different action-consequence trajectories
- (iv) Merge planning cortex with some sort of agent or "actor model" that then gives the model the ability to act; eval and fine-tune

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39 4 Proposal #2: RL-augmented, agent-based compositional modelling for 40 large-scale population dynamics

41 First and foremost, continuing this current modelling project (abstract included):

42 Agent-based models are a unique class of model that allows for hierarchically stochastic modelling of
43 environments. However, many of the current state-of-the-art for agent-based models do not effectively leverage
44 innovations in machine learning (ML) and reinforcement learning (RL) when modelling different agent
45 behaviours. Furthermore, most agent-based models instantiate agents in an environment-agnostic manner,
46 meaning agent behaviours are not affected by external factors from the environment. As a result, we tend to
47 see off-policy stochasticity in the behaviours of the agents, meaning the expressiveness and interpretability of
48 the models is limited. In order to remedy this, we propose an alternative agent-based modelling architecture
49 that makes use of a model-based RL learning process coupled with an internal hidden state that stores agent
50 actions, environmental stimuli, and their consequences. This allows us to define more realistic, learned
51 behaviours for individual agents and to allow for dynamic understanding of actions and consequences within
52 these environments. To test this hypothesis, we introduce two RL-augmented ABM environments and compare
53 them against NetLOGO and real-world data baselines. From this, we seek to conclude that RL-augmented
54 ABM and internal dynamics modelling for agents produces outsize returns for the accuracy and performance of
55 general ABM systems.

56 **Code:** <https://github.com/itsjackfan/online-agent-modelling>
57

58 **Immediate next steps:** Train a small, model-based RL agent instance that can be used to model behaviours
59 within 2 defined, Gym-like environments and investigate the feasibility of an individualised, RL-based dynamics
60 modelling approach; we define two baseline real-world environments that we want to examine: a standard
61 predator-prey model and a SIR-style modelling task (e.g. model COVID, or the 1918 Spanish flu pandemic)

62 **Follow-on:** using this data to create a multiscale latent representation of world dynamics + relevant dataset;
63 specifically, assuming this hypothesis holds and RL-augmented ABMs work well for modelling population
64 dynamics, we'll be able to construct a generalised set of trajectories for actions and consequences within
65 predator-prey or SIR-style models, which could potentially inform direct post-training of LLMs or training of
66 the action-consequence model architecture discussed above